Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	**Development of an AI speaker device and sLLM based services**			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	AI			
	Medium Category	Small Category	Medium Category	Small Category
	Healthcare service platform	*Silver Care Service Using sLLM AI and AI Speakers*		
Concept of Proposed Technology	**Core Concept** ● Development of a generative AI senior care solution based on sLLM This research and development aims to develop a lightweight large-scale language model (sLLM) optimized for senior care. Based on this, a generative senior care service platform for use in users' homes will be implemented, offering comprehensive services for seniors' safety, health, and emotional well-being. This will enhance their quality of life while reducing the social burden on families, caregivers, and the medical/welfare system. ● Development method 1. Development of a lightweight language model (sLLM) specialized for senior care: Developing a specialized sLLM AI model capable of supporting customized conversation models, multi-modality inference models, and content generation models for senior care.			

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	2. Home hub module hardware development: Developing a home hub module product that directly receives user voice commands within the user's home or connects to connected CCTV, Bluetooth speakers, etc. to input and output various audio, image, and video information, and connects to a central server system via the Internet. 3. Development of a generative senior care application: Integrating the home hub module into the senior care sLLM model as a base model, we developed various generative AI (Generative AI) service types, such as "safety management services," "life health assistance services," and "lifestyle value-added services."
Rationale for Joint R&D Project	**Critical Technical Barriers** Existing commercial large language models are subject to structural limitations that hinder their practical adoption in senior care services. These models rely on external cloud-based APIs, which incur continuous service costs and raise concerns related to data security, privacy protection, and policy compliance. In addition, their dependence on proprietary platforms limits system flexibility and long-term scalability. Open-source base models, while offering greater accessibility, are primarily designed for general-purpose applications and lack domain-specific optimization. As a result, they often produce insufficiently specialized inference outcomes in senior care scenarios and require excessive computational resources, making them unsuitable for deployment in resource-constrained edge or home environments. Furthermore, current senior care platforms face significant barriers in hardware integration and system deployment. Existing home monitoring and hub devices are often complex and costly, creating obstacles to large-scale adoption, particularly among elderly users. The lack of standardized, low-cost, and lightweight edge devices limits seamless integration with AI-based care services. In addition, many existing systems do not effectively support multimodal data processing and real-time adaptive inference, restricting their ability to respond accurately to diverse user contexts and emergency situations. These technological and infrastructural barriers impede the development of scalable, reliable, and user-centered senior care solutions, underscoring the need for optimized models, simplified hardware architectures, and integrated system design. The AI S/W development team, the H/W development team, and the team that understands the silver care business all need to collaborate with experts from different fields to carry out the project, and the school's research institute needs to participate to produce specialized technical products. In addition, a demand organization is needed to carry out global business. **Why Global Collaboration?** Purdue University maintains a strong foundation in human-centered artificial intelligence through specialized centers such as the Institute for Physical Artificial Intelligence (IPAI), Applied AI Research Center (AARC), and Research Computing and Data Analytics (RCAC). These institutes conduct systematic research on AI technologies related to human behavior, speech, perception, and decision-making, and provide an integrated ecosystem that connects fundamental research with system-level implementation and real-world deployment.

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	As a major regional hub university in the United States, Purdue is well positioned to identify and respond to local and national AI demands across industrial and public sectors. Moreover, its extensive experience in transforming background intellectual property into foreground intellectual property enables effective benchmarking and strategic technology internalization. Through collaboration with Purdue University, partners can enhance the commercialization potential of research outcomes and strengthen global competitiveness, making such cooperation a highly strategic and value-driven partnership.
Key Words	**1.sLLM** **2.RAG** **3.A.I Speaker** **4.Healthcare** **5.Conversational Voice Technology**

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	**Start of the Project**	Level (5)	**End of the Project**	Level (7)
R&D Objectives	**R&D Objectives** The goal of this research and development is to develop a lightweight large-scale language model (sLLM) optimized for senior care. Based on this, we aim to implement a variety of generative senior care services that can be utilized in users' homes. Through this, we aim to develop an integrated solution that comprehensively provides various lifestyle value-added services, including safety management and physical and emotional well-being management for senior users. **Detailed Objectives** 1) Development of a lightweight language model (sLLM) specialized for senior care - Open-source LLM model selection and parameter optimization: Currently commercialized models like Open AI's ChatGPT incur ongoing service fees and face security and other policy governance constraints. Therefore, this project developed a model based on Gemma3, the latest LLM open-source technology developed by Google DeepMind. (It is freely available for commercial use, offers the highest memory efficiency when using GPUs among recently announced LLMs, and offers excellent support for Korean and other languages, with a large parameter size of 27B.) We will evaluate our own model performance against a set of LLM models (utilizing existing senior care-specific conversation datasets, including language understanding, generation, and inference capabilities). We will then select a model with optimal parameters (model size) and develop our solution based on that model. - sLLM model development and fine-tuning: The open-source basic LLM model is not specialized for specific use cases and is useful for most general-purpose applications. Therefore, if the base model			

is used as is, it is difficult to derive specialized inference results in specialized scenarios, and inference consumes excessive computing resources. In this research and development, we plan to fine-tune the open-source LLM model by partially learning/training it with our own specialized senior care dataset at the application programming layer to isolate and lighten the model. Through this, we aim to maximize the inference performance of the sLLM base model itself and derive inference results optimized for the senior care field with optimal parameters and computing resource consumption.

a) Dataset Preparation: Preprocess our own dataset related to the senior care domain. Prepare training and validation sets.
b) Model Fine-Tuning: Apply a minimized learning rate, freeze some layers, and then perform step-by-step training.
c) Validation Evaluation: Evaluate model performance using the validation dataset. Monitor validation loss and accuracy to prevent overfitting.
d) Hyperparameter Tuning: Achieve optimal performance through learning rate, batch size, regularization, etc.
e) Testing/Deployment: Deploy STG testing and operational service. Compile real-user usability evaluations.

Development of a detailed scenario-specific inference model: To implement the senior care application targeted in this research and development, a scenario-specific inference/embedding model was developed that can perform required instructions, conduct conversations such as question and answering, and analyze and summarize situations/contexts when unstructured text input, conversational voice, images/videos, and structured/unstructured data are input from users or the environment. This process includes vector embedding of individual user structured/unstructured data, development of a RAG agent, and prompt engineering for individual inference models. This development area is directly connected to the development of a generative senior care application that will be carried out in the subsequent stage, and the scope of development is also a development area that intersects with each other at the system integration layer.

2) Home Hub Module HW Prototype Development:
- Home Hub Module Overview: This project will develop a prototype of a home hub module that can receive user voice input or image information from a connected camera within the user's home, output audio via a Bluetooth speaker, and connect to the central server system of a senior care solution via an internet connection. The prototype will utilize the Raspberry Pi development board. Upon commercialization, the company aims to expand the input/output interface to ensure compatibility with commercially available IP cameras and Bluetooth speakers. This minimizes the scope of its functionality, streamlining mass production and operating costs. This also minimizes the cost burden on users when deploying the product in their homes. The development goal is to develop a working prototype capable of piloting services by integrating it with the entire solution (server system, IP camera, BLE speaker).

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	3) Generative Senior Care Application Development: - Safety Management Service: When a user requests emergency assistance via voice or a user crisis is detected via a connected web camera, this service immediately analyzes the situation, takes immediate response measures, or takes additional generative measures to collect more accurate situational/contextual information. This process is an application software development process that integrates the input/output interfaces for inference based on the scenario-specific inference model developed in the previous step, and integrates and develops actual follow-up services (emergency dispatch requests, control system reports, and additional situational information collection) based on the inference results into our solution.
R&D Target Specifications	The verification of the research and development results aims to measure indicators such as "Does the developed AI service perform contextual analysis of the intended safety issue situation?" and "Does it perform Q&A using RAG accurately and reliably?" 1) Does the developed AI service perform contextual analysis of the intended safety issue situation? ● - Image Detection > Safety Issue Classification Accuracy ● Crisis Situation Classification Accuracy: (TP+TN)/(TP+FN+FP+TN) >= 95% ● - Voice Conversation > Safety Issue Classification Accuracy Crisis Situation Classification Accuracy: (TP+TN)/(TP+FN+FP+TN) >= 95% 2) Does the developed AI service perform Q&A using RAG accurately and reliably? - Faithfulness of conversational question-answering via RAG ● Faithfulness: Number of inferable information from the context (searched document) / Number of information included in the answer >= 95% ● - Context Recall of conversational question-answering via RAG ● Context Recall: Number of ground truth information inferable from the context / Number of ground truth information >= 95% ● - Speed of implementation of conversational question-answering service via RAG ● Average time from the time of voice query to the output of a voice response to the home hub module
R&D Plan	**Year 1**: Securing Core Foundational Technologies (Korean SME Partner) ● Development of a specialized sLLM senior care model ● Through the development of sLLM and RAG technologies, we are developing server software capable of conversational interaction using generative AI. Based on a GPU server, we will develop a variety of conversational scenarios. We will fine-tune the sLLM software to enable emotional and health-related conversations. We will also develop a custom RAG that can integrate various data sources and interface with external systems. Specifically, we will collaborate with insurance companies to develop an application that

	can support customers by linking with their internal systems. (Purdue University) ● Development of a voice recognition-based AI-dedicated speaker ● We are developing a compact AI speaker for home use that can perform various voice-based conversational functions for senior care. The AI speaker must have built-in Wi-Fi for voice recognition and server communication. Bluetooth functionality will also be used to enable app-based settings. An IP camera will be installed to enable behavioral recognition for senior care. **Year 2**: Platform Development and Data Expansion (Korean SME Partner) ● We train a large and diverse amount of data that sLLM and RAG can operate well, organize it into a vector DB to enable rich answers, and implement a service that allows Sw to operate well in real time by linking it with the health insurance system and the insurance company's internal system. (Purdue University) ● We develop stable features to ensure the device's smooth operation and work to improve voice recognition accuracy. We also produce prototypes to enable cost-effective mass production. Year 3: Reliability Validation and Commercialization (Korean SME Partner) ● We will work with New York City partners and US insurance companies to enable a demonstration project, and conduct small-scale commercialization work targeting actual users to check for technical shortcomings. (Purdue University) ● We are improving the sound sensor function and developing a commercially viable product so that the device can recognize various sounds that AI can recognize, such as changes in voice according to the user's emotions and sound detection for abnormal behavior of silverfish.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	(Purdue University) - Development of voice recognition technology for sound - Development of sound recognition technology for emotional and abnormal symptoms - Development of a commercially viable AI speaker - Development of a communication module that can be integrated with AI systems
Expected R&R of Korean SME	(Korean SME Partner) - Development of a conversational AI system based on sLLM and RAG

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Partner	- Construction and training of a conversational dataset - Development of an AI server agent platform - Development of a custom mobile app - Development of a voice recognition-enabled TTS/STT module - Construction of a cloud-based data analysis system

5. Expected Outcomes/Impact

Expected Outcomes and Impact	*1.* Technical Outcomes: Establishing a globally deployable hyperspectral AI analytics framework and standardization basis ● - AI Technology Advancement: The development of generative AI services based on sLLM will accelerate the advancement of core AI technologies such as natural language processing and computer vision. In particular, the commercialization of care and monitoring services based on conversations with actual users will contribute to the continuous production of conversational learning sets, thereby enhancing the comprehension and contextual analysis capabilities of AI models. ● - Data-Driven Research Advancement: Research based on diverse conversation and image data collected from elderly users can be activated. This will provide new insights into academic research analyzing health, lifestyle habits, and psychological states, and can serve as a data asset for future academic advancements in the fields of elderly welfare and health care. ● - Personalized Medical and Care Service Research: Care services utilizing sLLM increase the potential for personalized medical and care services tailored to individual user characteristics. This will play a crucial role in the future development of personalized healthcare technologies and preventive healthcare research and services. ● - Promoting Interdisciplinary Research: Research that integrates diverse disciplines such as AI, health care, ergonomics, and psychology will become more active. This is expected to contribute to the advancement of science and technology by enabling the development of new convergence technologies and services. *2.* Economic/Industrial Impact: Commercialization of an export-oriented product/platform and scalable industrial applicability ● Market Expansion and Growth: As the population ages, demand for related industries is expected to surge due to the increasing elderly population. SLLM-based safety monitoring and health care services can be innovative solutions to meet the growing demand for care services in this aging society, driving innovation and growth in related industries and services.

	● New Market Creation: The US is already witnessing rapid growth in related industries, with a variety of non-medical startups based on the WHO's Social Determinants of Health (SDoH) initiative. In Korea, even seniors living in remote areas and urban areas have a high demand for health-related services in their daily lives. Therefore, our company's goal of creating a new market for concierge services based on lifestyle health data is highly anticipated. *3.* Societal Impact: Reduced fresh-produce waste and improved farm income stability ● Reducing Social Burden: The burden of social care costs resulting from the explosive growth of the aging population and the decline in the young working-age population due to low birth rates is expected to become a burden for the entire nation and society. Therefore, safety monitoring and care services utilizing sLLM-based generative AI will contribute to reducing the national social welfare budget in the long term. ● - New Job Creation: The growing demand for new technical personnel required to develop, maintain, and manage AI services will lead to the creation of new technical jobs. Furthermore, the development of various value-added businesses and related industries, such as lifestyle concierges, which design customized care services for seniors, is expected to contribute to new business opportunities and job creation.